

Department of Computer Science
MCA SEM-II
Subject: Artificial Intelligence
Assignment – II

INSTRUCTIONS:

- You are expected to provide detailed responses to each question.
 - Remember to make an honest attempt.
 - The grading will be related to the conceptual understanding and the level of explanation which you will provide in your assignment.
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1. What do you mean by State Space? Formally define it and explain it with the help of a classic puzzle (like the 8-puzzle or Water Jug).
2. Explain the Production System model in AI. Describe its four basic components: a set of rules, a knowledge base, a control strategy, and a rule applier.
3. AI problems possess different Problem Characteristics (e.g., Is the problem decomposable? Can solution steps be ignored or undone?). List and briefly explain five major characteristics.
4. Take any two problem characteristics of your choice and provide a detailed reasoning of how they apply to Chess versus Theorem Proving.
5. Explain the Breadth-First Search (BFS) algorithm. Provide a step-by-step trace using a small graph and explain why it is considered "optimal" for unweighted graphs.
6. Explain the Depth-First Search (DFS) algorithm. Provide a trace and discuss its space complexity advantages over BFS.
7. BFS guarantees the shortest path but is extremely memory-expensive. Describe a real-world scenario where BFS would likely fail due to hardware limitations.
8. DFS may get stuck in infinite loops or deep paths that don't lead to a goal. How can we modify DFS (e.g., Depth-Limited Search or Iterative Deepening) to avoid these issues?
9. Compare Discrete vs. Continuous State Spaces. Provide one real-life example of each and explain why search techniques differ for them.
10. What is a Heuristic function? Why is it often called "informed search," and how does it differ from a "blind search"?
11. Explain the Generate-and-Test method. Is it an efficient strategy for complex problems like Chess? Why or why not?
12. Explain Simple Hill Climbing and Steepest Ascent Hill Climbing. What is the fundamental difference in how they choose the next move?
13. Hill Climbing is prone to the Local Maxima problem. Explain this problem and describe one strategy (like Random Restart or Simulated Annealing) used to overcome it.

14. In the context of Hill Climbing, explain the Plateau and Ridge problems. Provide a real-time application where a "Ridge" might make optimization difficult.
15. If you had to solve the Water Jug problem without using BFS or DFS, what heuristic-based strategy would you design? Define your heuristic function.
16. Explain the Best First Search algorithm. Show with an example how it uses the evaluation function $f(n)=h(n)$
17. Best First Search can be "greedy." Describe a situation or a graph structure where a Greedy Best First Search would fail to find the shortest path.
18. Define a Constraint Satisfaction Problem (CSP) using the (V,D,C) framework (Variables, Domains, and Constraints). Use a Map Coloring example to illustrate.
19. Why is Sudoku classified as a single-agent discrete CSP, whereas Chess is classified as a multi-agent discrete state space search? Explain the difference in their constraints.
20. Explain the *A Algorithm** in detail. Define the formula $f(n)=g(n)+h(n)$ and explain what each component represents.